

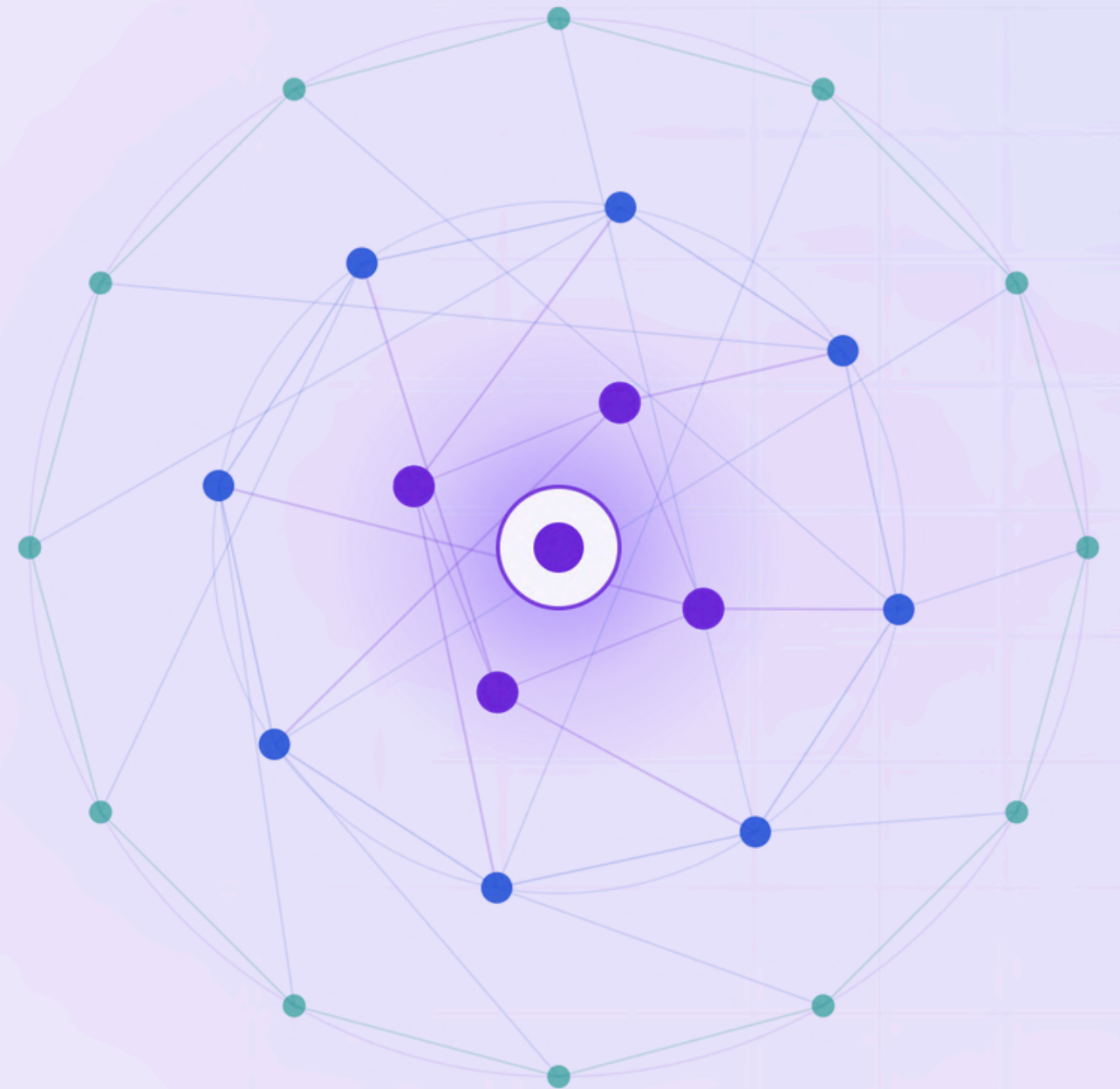
SAMSUNG PRISM

Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - **Theme 2 · Smart Guided Troubleshooting Engine**
- Team Name - **Thapar_Nexora_2**
- College Name - **Thapar Institute of Engineering and Technology , Patiala**
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- Submission Github link -

https://github.com/khushichhabra7921/PRISM_GENAI_HACKATHON_Y2026_UPDATED_GITHUB



 <300 ms on known issues

 <8 s on new issues

 0 invented steps

 0 invalid deeplinks

Theme

Theme 2 · Troubleshooting: turn vague Galaxy complaints into deeplinked, one-tap troubleshooting plans.

People describe **symptoms**.
Settings speak in **menus**.

DISPLAY · BATTERY

"Screen flickers and the battery dies fast"

PERFORMANCE

"My phone got slow after the update"

NAVIGATION

"Swipe gestures go the wrong way after installing an app"

TODAY

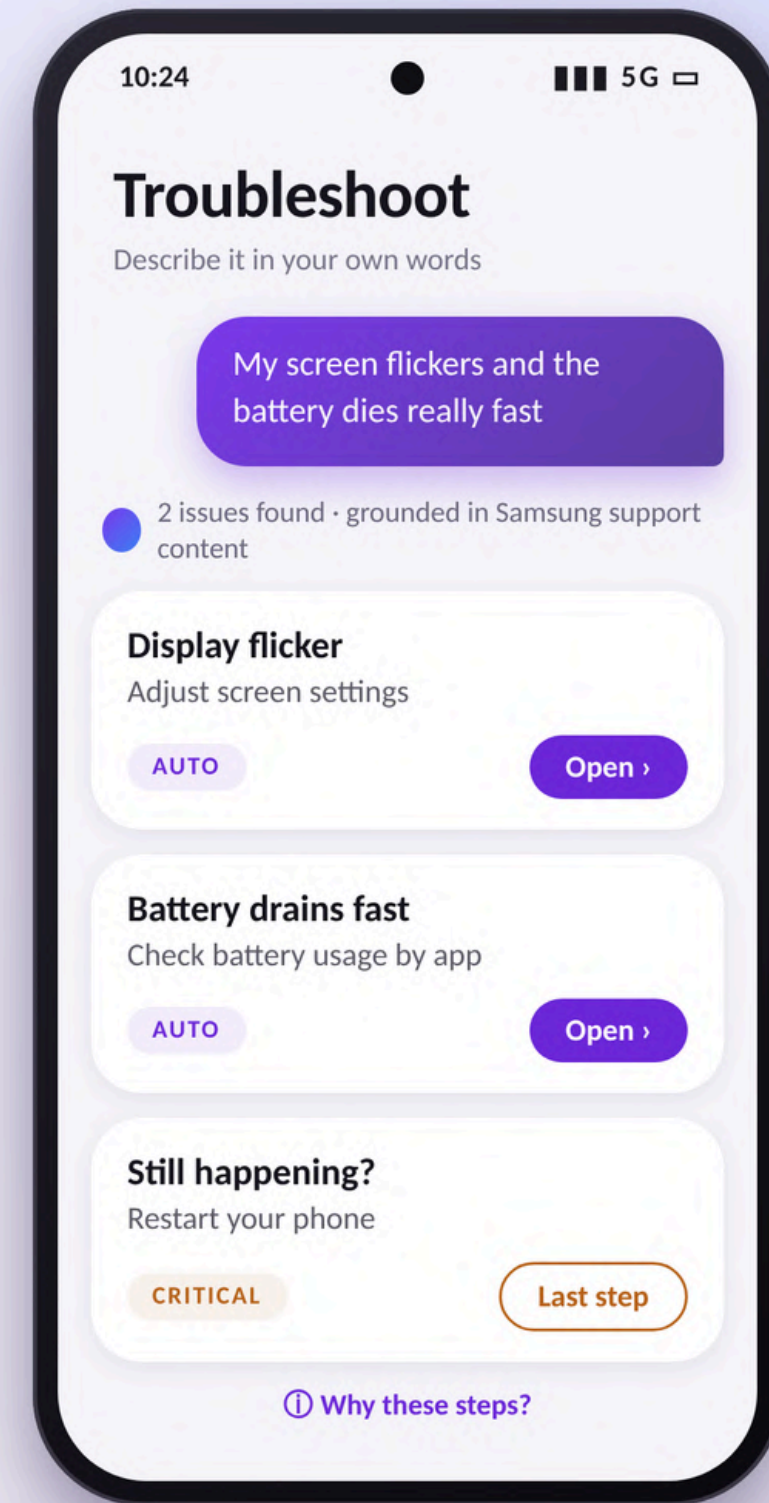
~15 min

manual triage per scenario: agent reads SIIS, orders steps, user hunts through Settings

OUR TARGET

<300 ms

known issues via semantic cache; < 8 s for new ones, fully grounded



Existing Solutions & Gaps

Each product owns part of the job. None turns a vague complaint into a grounded, ordered, validated plan with the exact Settings screen.

Solution	Good at	Natural language	Trouble-shooting	Ordered plan	Exact deeplink	Grounded + validated	Gap vs. our target workflow
Samsung Bixby	natural-language assistant; runs device actions	Core	Partial	—	Partial	—	executes actions; doesn't plan from grounded knowledge
Device Care	battery, storage, memory and device health	Partial	Partial	Partial	Partial	—	diagnoses state; no complaint → ordered plan
Samsung Members	diagnostics, support content, error reports	Partial	Core	Partial	Partial	—	user still has to find the right path
Warranty & Care	Bixby help, symptom search, self-diagnostics	Core	Core	Partial	Partial	—	no plan validation or deeplink layer
Samsung Help Center	deep library of issue-specific articles	Partial	Core	—	—	—	retrieval only; user turns text into taps
Google Pixel Help	NL help, device health, diagnostics	Core	Core	Partial	Partial	—	no Samsung SIIS grounding, deeplinks or cache
Microsoft Get Help	guided troubleshooters and diagnostics	Core	Core	Core	Partial	—	built for Windows, not Galaxy Settings
FAQ / web search	broad coverage, easy to find	Partial	Core	—	—	—	user interprets, orders and executes
Human / remote support	expert diagnosis for unusual cases	Core	Core	Core	—	—	slow and costly; not instant or deterministic
Our engine	complaint → verified one-tap plan	✓ Core	✓ Core	✓ Core	✓ Core	✓ Core	the full chain, with an evidence trail



Assistants act...
but don't plan from trusted knowledge



Diagnostics measure...
but don't explain what to do next

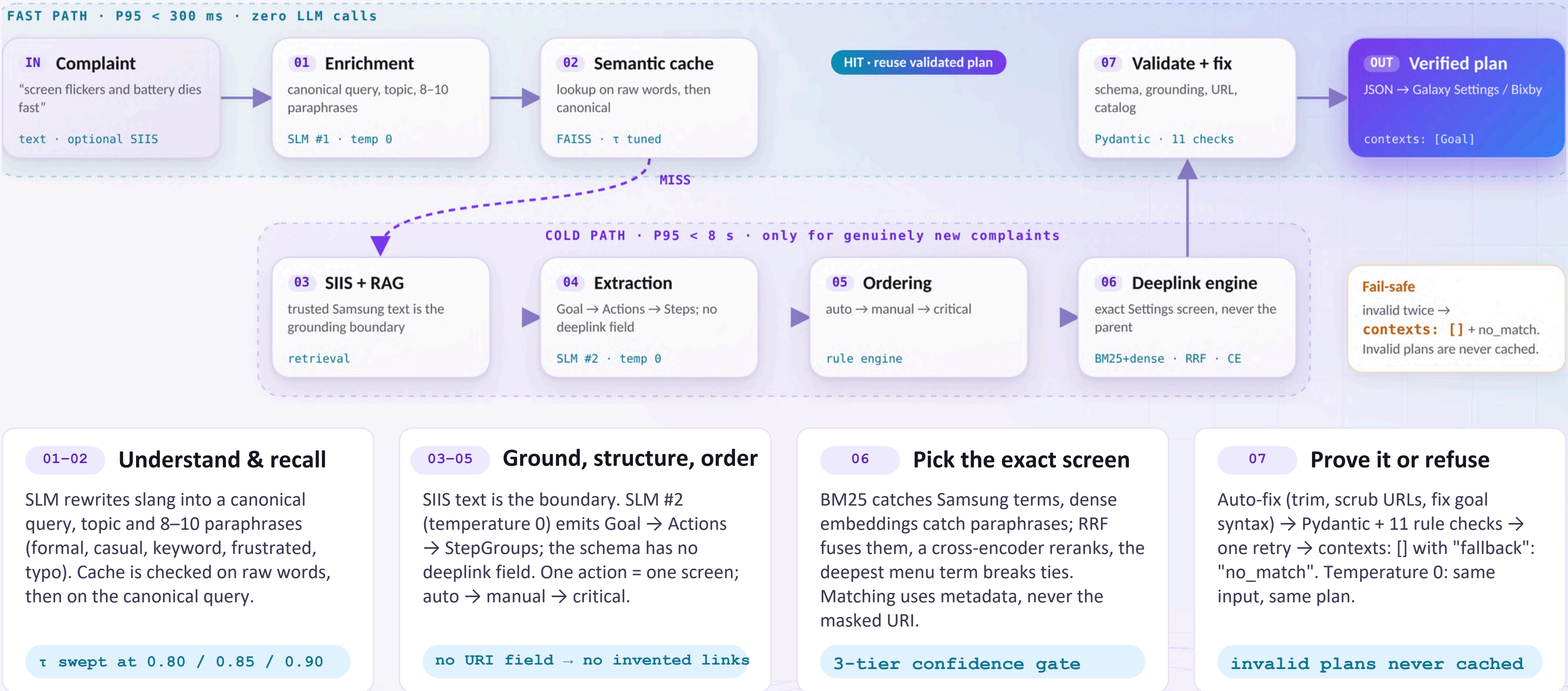


Articles explain...
but leave every tap to the user

We connect intent → knowledge → action.
Bixby stays the execution layer.

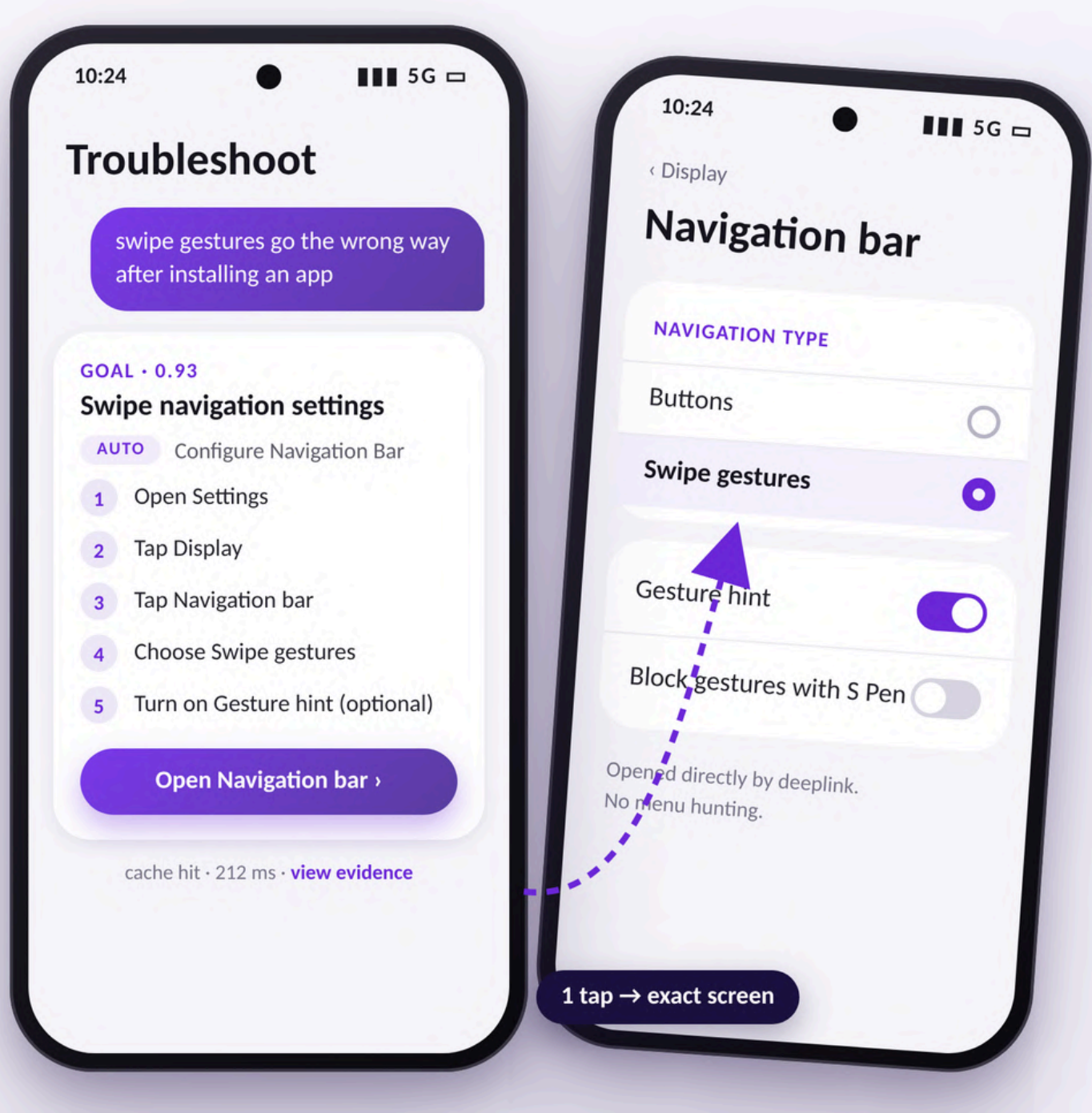
Our Solution & Architecture Diagram

LLMs reason, retrieval grounds, deterministic code validates. Only genuinely new complaints ever pay for reasoning.



Demo & Product Walkthrough

A real example from the brief, end to end: what the user sees, what the API returns, and the receipt behind every action.



```
sgte · /v1/troubleshoot 200 · 212 ms

# request
POST /v1/troubleshoot
{ "query": "phone swipe gestures wrong direction after app install" }

# response · 200 OK
{ "contexts": [{
  "goal": "Follow these steps to perform this Swipe Navigation Troubleshooting",
  "title": "Swipe navigation settings",
  "score": 0.93,
  "actions": [{
    "actionName": "Configure Navigation Bar Settings",
    "description": "It will let you choose navigation type",
    "category": "auto",
    "stepGroups": [{ "steps": [
      "Navigate to and open Settings.",
      "Tap on Display.", "Tap on Navigation bar.", ...],
      "actionableDeeplink": {
        "deeplink": "bixby://dummy_positive" }
    ] } ] },
  "meta": { "latency_ms": 212, "cache_hit": true,
    "model": "...", "cost_usd": 0.0 }
}

# optional receipt, outside the graded schema
"evidence": { "source_sentence": "...",
  "bm25", "dense", "rerank", "margin", "tier" }
```

- 1 **Vague complaint in**
"swipe gestures go the wrong way after installing an app"
- 2 **Understood**
enrichment maps it to gesture-navigation settings · topic Display
- 3 **Recalled**
a paraphrase hits a validated plan: 212 ms, \$0.00, no LLM call
- 4 **Planned**
one auto action, five grounded steps, least disruptive first
- 5 **One tap**
deeplink opens Display › Navigation bar; evidence on request

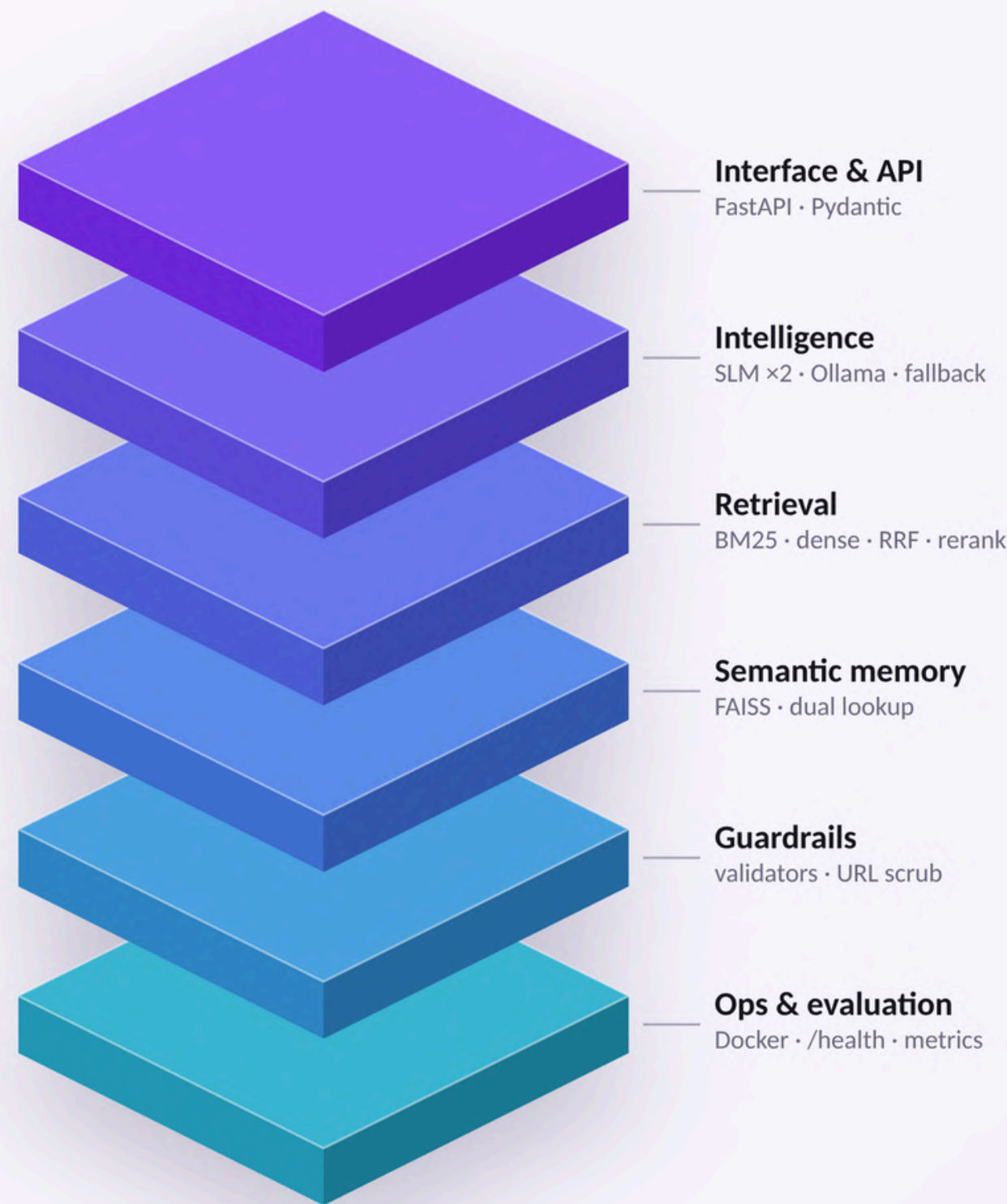
EVIDENCE TRAIL · PER ACTION

● source sentence	● bm25 score
● dense score	● rrf rank
● rerank score	● runner-up margin
● confidence tier	● final decision

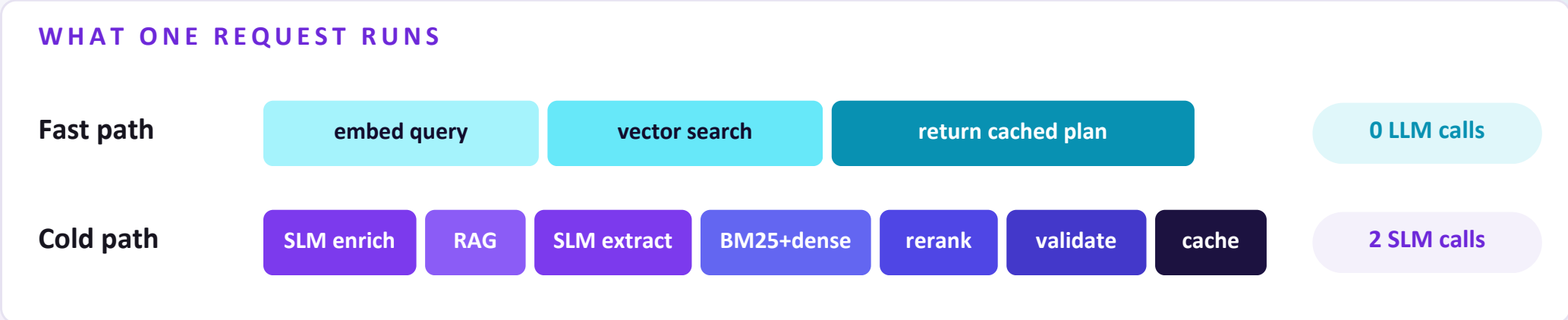
Demo video. : <https://youtu.be/UWmbvZDs3Ao?si=QjdoR6tNjP7mBOxH>

Tools and Tech Stack Used

A layered, containerised, SLM-first stack. Every layer is there for a latency or safety reason.



Layer	What we run	Why this, not the obvious alternative
Interface	FastAPI · Pydantic · schema.py	contract enforced at the boundary; JSON only
Intelligence	local SLM (Ollama) + hosted fallback · llm_client.py · temp 0	two narrow jobs don't need a frontier model per call
Retrieval	SIIS RAG · BM25 · dense · RRF · cross-encoder	keywords hit parent menus; embeddings miss exact terms
Memory	FAISS · dual lookup (raw + canonical) · pre-warmed	exact-string keys miss paraphrases
Guardrails	Pydantic validators · regex URL scrub · fuzzy grounding	prompts can't be trusted with word counts
Ops & eval	Docker Compose · /health · benchmark · metrics.md	numbers logged per request, not estimated



Impact & Use Case

Every Galaxy shipped is a future support conversation. Here is who benefits, and how the engine behaves in three real situations.

241.2M

Samsung smartphones shipped in 2025
~19% global share

~15 min

manual triage per scenario today
problem brief

61%

prefer self-service for simple issues
Salesforce, 2025

92%

would use a self-serve knowledge base
Higher Logic, 2024

HOW WE SIZE IT
time saved $\approx$
automation rate $\times$
15 min
 $\times$ query volume
rate = measured cache-hit rate

A Known issue, new wording
FAST PATH

- “My Wi-Fi keeps dropping”
- paraphrase hits the dual cache
- validated plan, zero LLM calls
- one tap to the exact Wi-Fi screen

B Two problems, one sentence
MULTI-SYMPTOM

- “Screen flickers and my battery dies”
- split into Display + Battery
- each runs the full grounded pipeline
- two verified goals in one response

C An issue SIIS doesn’t cover
SAFE FAILURE

- complaint with no source text
- contexts: [] + no_match, nothing invented
- clustered offline with similar misses
- gap report for the SIIS content team

 Users

one tap instead of menu hunting; slang and typos understood; both halves of a complaint answered

best for older & first-time users

 Support teams

escalations arrive with the evidence trail; no-match clusters show where SIIS is thin

5,255 courses · 39,703 trainees in 2025

 Environment

fewer service visits for config-only issues; a cache hit makes zero LLM calls

SLM-first, proven by ablation

 Samsung

demonstrable AI safety; pattern ports to SmartThings & Galaxy Watch

3,000+ touchpoints · 12,500+ engineers (India)

Innovation Highlights, Results and Limitations

Four capabilities no public build of this theme we reviewed attempts, the targets we are held to, and where it can still fail.



01
Evidence trail

Every action can carry its SIIS source sentence, BM25 / dense / rerank scores and why the runner-up lost.

grounding you can show



02
Multi-symptom split

Two problems in one sentence become two goals, each run through the full grounded pipeline.

no half-answered complaints



03
No-match clustering

Unanswerable queries are clustered offline, so gaps in SIIS coverage surface as a work list.

failure becomes a signal



04
Graded confidence

Three tiers instead of match-or-nothing: auto-attach, attach with a visible score, or fall back.

more coverage, same precision

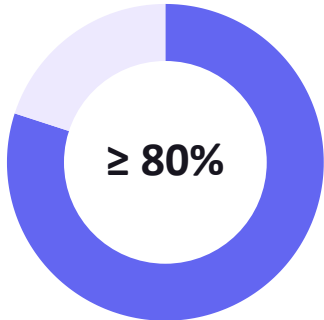
RESULTS · TARGETS WE ARE HELD TO



≥ 99%

schema-valid output

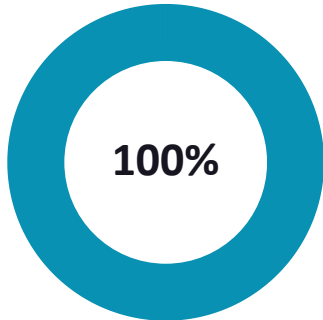
P95 ≤ 300 ms cached



≥ 80%

paraphrase cache hits

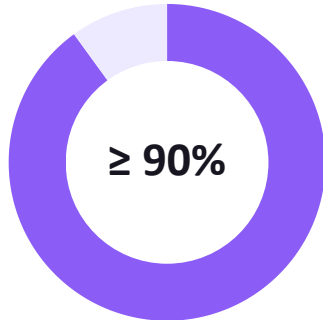
P95 ≤ 8 s cold



100%

catalog-valid deeplinks

0 URL leaks



≥ 90%

auto actions with deeplink

Measured values go in metrics.md (N ≥ 30 per path) and are reported even if they miss.

LIMITATIONS → MITIGATIONS

Cache hit may land < 80%	→ dual lookup + honest τ sweep
Keyword match stops at parent menu	→ hybrid + rerank + deepest-menu tie-break
Near-tied deeplink candidates	→ graded gate drops rather than guesses
Answers bounded by SIIS coverage	→ no_match is correct; clusters flag gaps
Exact-span grounding too strict	→ fuzzy / semantic step matching
SLM quality · English-first	→ hosted fallback; more languages next

Smart Guided Troubleshooting Engine | Samsung PRISM GenAI Hackathon 2026 | Theme 2

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What's Next

Where the architecture goes from here: broader coverage, a system that learns from use, and reach beyond the phone.

- **Broader device and settings coverage:** extend the deeplink index across more Galaxy models, One UI versions and settings paths. Every new deeplink passes the 11 validators before it is served, so users are never sent to a screen that does not exist on their device.
- **Confirmed-fix learning loop:** when a user confirms a step solved their problem, that signal is logged and fed back to retrain ranking and tune the match threshold τ . Proven fixes rise to the top, and τ is set from real outcomes rather than guesswork.
- **No-match clusters feed SIIS content:** queries with no confident match are grouped by intent, and the largest clusters are ranked by volume and handed to content authors as new guides. Every miss becomes a concrete content gap.
- **More query languages and registers:** add further languages and cover formal, casual and misspelled phrasing, keeping ranking quality consistent so users get the same accurate guidance however they describe the problem.
- **Hardened offline / on-device SLM:** run the core troubleshooting flow on the device without a connection, with a graceful fallback to the cloud path when a query needs it. Guidance stays available with weak or no network.
- **Cache that improves with use:** the semantic cache is pre-warmed from real query patterns, so frequent questions are served from cache and repeat queries stay under 300 ms even as traffic grows.

Brownie Points (Differentiation)

Every differentiator is backed by published evidence, not internal claims. Our own measured numbers go in metrics.md.

61%

of users prefer self-service over an agent for simple issues *Salesforce, 2025*

92%

would use a self-serve knowledge base if one were offered *Higher Logic, 2024*

70%

of enterprise AI tools expected to use RAG by end of 2025 *Gartner*



Sub-300 ms hits, sub-8 s cold

known issues never wait on an LLM

caching: up to 90% lower LLM cost, up to 250x lower latency (Redis / Percona, 2025)



Two-tier paraphrase cache

raw words, then the canonical query

61–69% of repeat queries served without a fresh LLM call (GPTCache, arXiv:2411.05276)



Hybrid retrieval, exact screens

BM25 + dense + RRF + cross-encoder

hybrid beats either method by 26–31% NDCG (BEIR aggregate)



Guardrails live in code

syntax, length, deeplinks, URL leaks

retrieval + validation: hallucinated steps ~21% → < 7.5% (NAACL 2024)



Fails safe, never invents

no source → contexts: [] + no_match

grounded generation: 70–90% less hallucination (RAG studies, 2025–26)



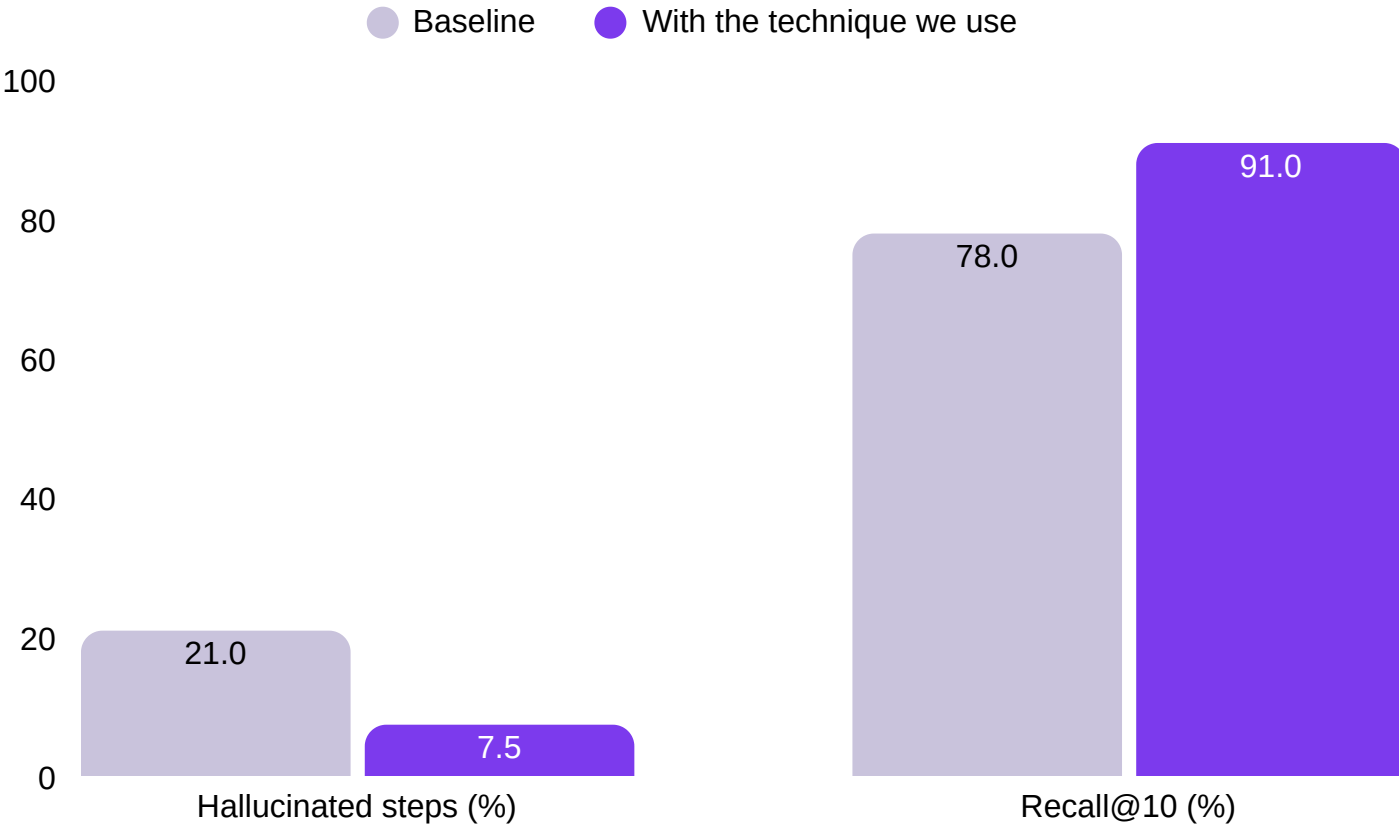
Tuned and measured

ablations, τ sweeps, false-hit rates

untuned fusion can trail a tuned baseline by ~10 MRR points (AIMultiple, 2025)

TWO RECEIPTS, CHARTED

Published results behind our design choices



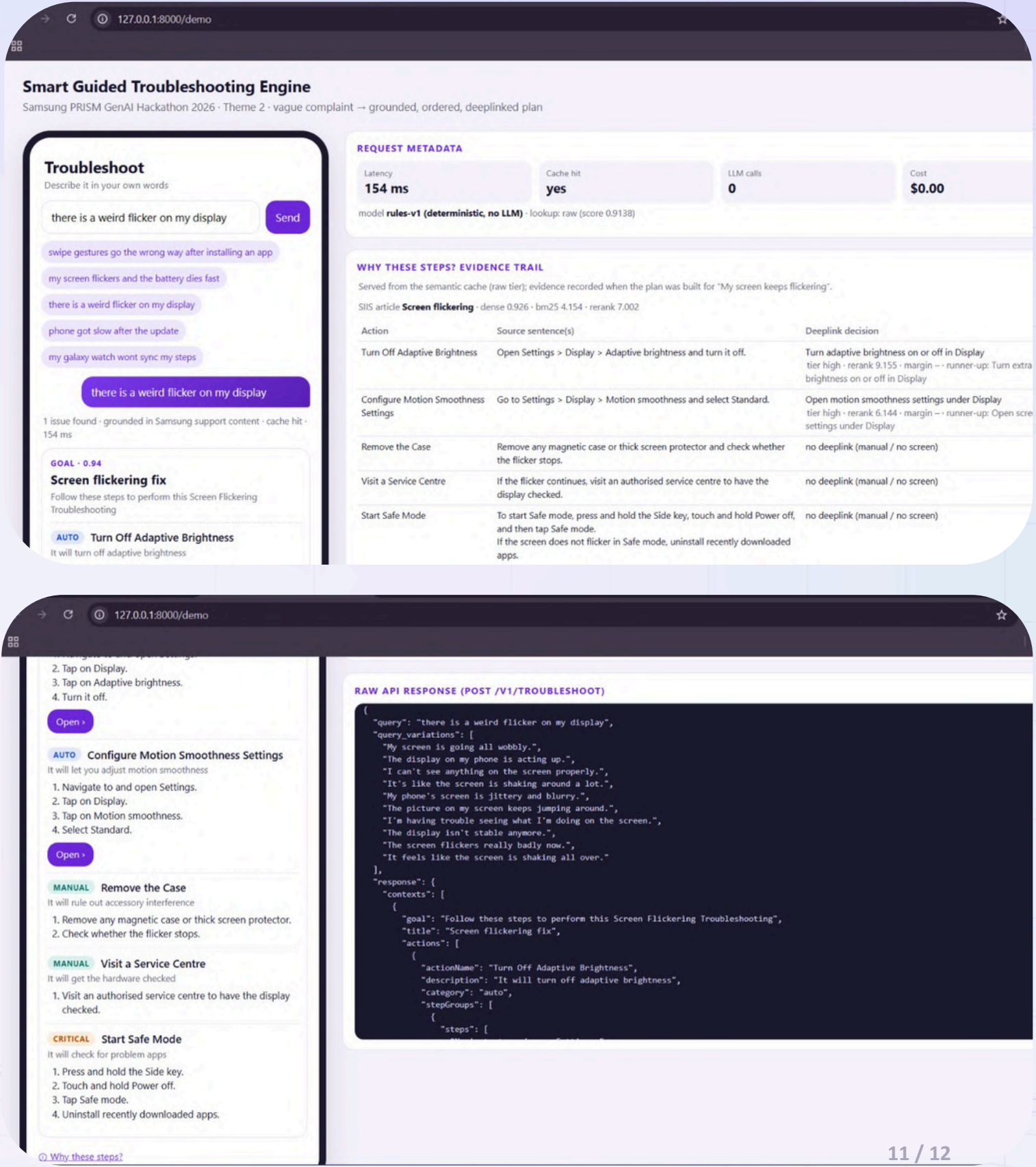
Left: ungrounded vs. retrieval + validation (Ayala & Bechard, NAACL 2024). Right: single retriever vs. hybrid (BEIR aggregate).

Checklist: Updated on Public GitHub

What judges open first. Mark each item and paste the links before the form goes in.

Deliverable	Status	Link / note
Working prototype code public or shared GitHub repo	Yes	https://github.com/khu-shichhabra7921/PRISM_GENAI_HACKATHON_Y2026_UPDATED_GITHUB
README reproducible setup instructions	Yes	<code>docker compose up</code>
Demo video max 5 minutes, YouTube or Drive	Yes	https://youtu.be/UWmbvZDs3Ao? <code>si=QjdoR6tNjP7mBOxH</code>
Presentation PPT or PDF	Yes	this deck

LIVE DEMO



Research and References

- **Luo, J., Li, T., Wu, D., Jenkin, M., Liu, S., & Dudek, G. (2024).** Hallucination Detection and Hallucination Mitigation: An Investigation. Samsung AI Center, Montreal. [arXiv:2401.08358](<https://arxiv.org/abs/2401.08358>)
- **RAG / Grounded Generation** Lewis, P., Perez, E., Piktus, A., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS 2020. [arXiv:2005.11401](<https://arxiv.org/abs/2005.11401>)
- **Shuster, K., Poff, S., Chen, M., Kiela, D., & Weston, J. (2021).** Retrieval Augmentation Reduces Hallucination in Conversation. [arXiv:2104.07567](<https://arxiv.org/abs/2104.07567>)
- **Nogueira, R. & Cho, K. (2019).** Passage Re-ranking with BERT. [arXiv:1901.04085](<https://arxiv.org/abs/1901.04085>) — the standard citation for cross-encoder reranking, justifying your rerank stage after RRF fusion.
- **Anonymous (2025). Winning Big with Small Models:** Knowledge Distillation vs. Self-Training for Reducing Hallucination in Product QA Agents. [arXiv:2502.19545](<https://arxiv.org/abs/2502.19545>)
- **Applied Domain — Troubleshooting via RAG** Khanda, R. (2024). Agentic AI-Driven Technical Troubleshooting for Enterprise Systems: A Novel Weighted Retrieval-Augmented Generation Paradigm. [arXiv:2412.12006]. (<https://arxiv.org/abs/2412.12006>)

Mentor : Shruti Aggarwal

Thank you

Understand

Ground

Plan

Navigate

Prove

Smart Guided Troubleshooting Engine Nexora · Thapar Institute of Engineering and Technology , Patiala

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India